

# Report - Electricity Bill Prediction

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**Project:** Supervised Learning - Regression models. **Date:** 2025-11-06 14:07:45 **Author:** Bruno Silva

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**Environment:**

- pandas: 2.3.1
  - NumPy: 2.3.1
  - scikit-learn: 1.7.2
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## 1. INTRODUCTION

### Project Objective

Develop a regression model to accurately predict monthly electricity bill amounts based on partial month consumption data, enabling proactive budget management and early cost forecasting.

### Business Problem

- **Early Budget Forecasting:** Predict final monthly bill using only partial month data (e.g., first 10, 15, 20, or 25 days)
- **Financial Planning:** Enable consumers to anticipate monthly costs before the billing cycle ends
- **Anomaly Detection:** Identify unusual consumption patterns or billing anomalies early in the month
- **Tariff Optimization:** Understand cost drivers and consumption patterns to optimize electricity usage
- **Budget Management:** Reduce uncertainty in household energy expenses through proactive forecasting

### Task Type

- **Supervised Regression**
- **Target:** `preco_fatura` (monthly electricity bill in €)
- **Challenge:** Predict full month bill using only partial month consumption data
- **Innovation:** Multi-scenario training with cutoff days (10, 15, 20, 25) to enable predictions at different points during the month

### Data Sources

Historical electricity consumption data with:

1. Daily meter readings (cumulative kWh for off-peak and peak periods)
  2. Portuguese electricity tariff structure (bi-hourly rates)
  3. Weather data (temperature, humidity, conditions)
  4. Temporal features (month, season, weekday, holidays)
  5. Engineered features for partial month scenarios
  6. Portuguese billing rules (network access, audiovisual contribution, taxes, VAT)
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## 2. EXPLORATORY DATA ANALYSIS (EDA)

### Dataset Summary

- **Total Records:** 184
- **Features:** 25
- **Training Samples:** 148
- **Test Samples:** 36

# Relatório de Análise Exploratória - Energy Coast Forecast

**Data de Geração:** 2025-11-06 14:06:59

## 1. Estatísticas Descritivas do Consumo (kWh)

| Estatística   | Valor      |
|---------------|------------|
| Contagem      | 46         |
| Média         | 352.35 kWh |
| Desvio Padrão | 41.20 kWh  |
| Mínimo        | 194.00 kWh |
| Q1 (25%)      | 332.25 kWh |
| Mediana (50%) | 352.96 kWh |
| Q3 (75%)      | 379.00 kWh |
| Máximo        | 447.00 kWh |
| IQR           | 46.75 kWh  |
| Skewness      | -0.986     |
| Curtose       | 3.985      |

## 2. Detecção de Outliers (Método de Tukey)

- **Limite Inferior:** 262.12 kWh
- **Limite Superior:** 449.12 kWh
- **Outliers Detectados:** 1 (2.2%)

## 3. Matriz de Correlações

| Variável  | Vazio kWh | Fora-Vazio kWh | Dias no Mês | Consumo Total |
|-----------|-----------|----------------|-------------|---------------|
| Vazio Kwh | 1.000     | 0.266          | 0.140       | 0.859         |

| Variável          | Vazio kWh | Fora-Vazio kWh | Dias no Mês | Consumo Total |
|-------------------|-----------|----------------|-------------|---------------|
| Fora Vazio Kwh    | 0.266     | 1.000          | -0.024      | 0.722         |
| Dias No Mes       | 0.140     | -0.024         | 1.000       | 0.088         |
| Total Consumption | 0.859     | 0.722          | 0.088       | 1.000         |

4. Distribuição por Estação

| Estação   | Contagem | Média (kWh) | Desvio Padrão (kWh) |
|-----------|----------|-------------|---------------------|
| Inverno   | 11.0     | 387.56      | 31.88               |
| Outono    | 11.0     | 331.67      | 15.20               |
| Primavera | 12.0     | 365.11      | 25.72               |
| Verão     | 12.0     | 326.26      | 50.60               |

5. Interpretação e Recomendações

5.1 Análise de Assimetria

- **Skewness = -0.986:** Distribuição assimétrica negativa (cauda à esquerda)
- **Implicação:** RMSE pode penalizar mais erros em valores baixos.

5.2 Impacto de Outliers

- **1 outliers detectados** podem influenciar modelos sensíveis
- **RMSE será mais penalizado** por estes valores extremos que MAE
- **Recomendação:** Considerar modelos robustos ou tratamento de outliers

5.3 Prevenção de Data Leakage

- **CRÍTICO:** Qualquer split treino/teste deve **respeitar a temporalidade**
- **Método recomendado:** Time Series Split ou separação por data
- **Evitar:** Random split que pode usar dados futuros para prever o passado

3. PREPROCESSING PIPELINE

Steps Executed

1. Data Integration

- Merged daily electricity readings with monthly billing data
- Integrated weather data (temperature, humidity) from Montijo station
- Aligned temporal features across all data sources

2. Partial Month Simulation

- Created scenarios for cutoff days: 10, 15, 20, and 25

- Accumulated consumption up to each cutoff day
- Simulated real-world prediction scenarios at different points in the month

### 3. Feature Engineering

- **Accumulated Consumption:** Total kWh consumed up to cutoff day (off-peak and peak)
- **Temporal Features:** Month, day of week, season (inverno, primavera, verão, outono)
- **Weather Features:** Average temperature and humidity up to cutoff day
- **Rolling Statistics:** 7-day rolling averages for consumption patterns
- **Tariff Features:** Days in month for proration of fixed costs

### 4. Target Variable Calculation

- Applied Portuguese electricity tariff rules (2024/2025)
- Computed bill components: energy costs (vazio/fora-vazio), network access, audiovisual contribution
- Applied special consumption tax and VAT (6% first 200 kWh, 23% above)
- Target: `preco_fatura` (total monthly bill in €)

### 5. Train-Test Split (80/20)

- Temporal validation: Earlier months for training, later months for testing
- Prevents data leakage from future to past
- Maintains realistic production scenario evaluation

### 6. Feature Transformation

- **StandardScaler:** Applied to numeric features (fitted on training data only)
- **OneHotEncoder:** Applied to categorical features (season, day of week)
- Drop='first' to avoid multicollinearity (dummy variable trap)
- All transformations fitted on training set only to prevent data leakage

### 7. Data Quality Checks

- Handled negative consumption values (set to zero, keep fixed charges)
- Validated billing calculations with detailed logging
- Ensured feature consistency between train and test sets

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## 4. MODELS TRAINED

### Algorithms Evaluated

#### Linear Regression (multiple)

- Training Time: 0.0030 seconds
- Notes: Standard linear regression with all features

#### SVR (linear kernel)

- Training Time: 0.0030 seconds
- Notes: Linear SVR - robust to outliers, epsilon-insensitive loss

**SVR (RBF kernel)**

- Training Time: 0.0043 seconds
- Notes: Non-linear SVR with RBF kernel - captures complex patterns

**Linear Baseline (1 feature)**

- Training Time: 0.0250 seconds
- Notes: Simplest possible model - uses only total accumulated consumption

**Lasso Regression**

- Training Time: 0.0640 seconds
- Notes: L1 regularization - automatic feature selection, sparse models

**Polynomial Regression (degree 2)**

- Training Time: 0.0660 seconds
- Notes: Non-linear model via polynomial expansion - HIGH RISK OF OVERFITTING!

**Random Forest**

- Training Time: 0.3760 seconds
- Notes: Ensemble of trees - excellent for non-linear patterns and feature importance

**Ridge Regression**

- Training Time: 4.8265 seconds
- Notes: L2 regularization - prevents overfitting with correlated features

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5. RESULTS AND PERFORMANCE METRICS

5.1 Complete Model Comparison

| Rank | Model           | RMSE    | MAE     | R <sup>2</sup> | Max_Error | Training_Time_s |
|------|-----------------|---------|---------|----------------|-----------|-----------------|
| 1    | svr_rbf         | 2.69463 | 2.08049 | 0.60633        | 5.62561   | nan             |
| 2    | svr_linear      | 3.04051 | 2.38385 | 0.498782       | 6.62411   | nan             |
| 3    | ridge           | 4.30768 | 3.58707 | -0.00605386    | 7.7256    | 4.8265          |
| 4    | linear_baseline | 4.3535  | 3.68964 | -0.0275729     | 9.89174   | 0.0249925       |
| 5    | linear_multiple | 4.43098 | 3.76619 | -0.064471      | 8.83383   | nan             |
| 6    | lasso           | 4.93987 | 4.26903 | -0.323016      | 8.17044   | 0.0640175       |
| 7    | random_forest   | 5.32391 | 4.15718 | -0.536724      | 13.8101   | 0.376034        |
| 8    | polynomial_deg2 | 35.0258 | 28.3079 | -65.5137       | 77.9562   | nan             |

 **Best Model: svr\_rbf**

- RMSE: 2.6946 €
- MAE: 2.0805 €
- R²: 0.606330
- Max Error: 5.63 €

5.2 Cross-Validation Results (Time Series Split)

| Model           | Mean_RMSE | Std_RMSE | Min_RMSE | Max_RMSE | CI_Lower | CI_Upper |
|-----------------|-----------|----------|----------|----------|----------|----------|
| svr_rbf         | 5.68777   | 1.28371  | 4.11969  | 7.77157  | 4.09384  | 7.2817   |
| random_forest   | 6.65917   | 1.99798  | 4.06411  | 8.9438   | 4.17835  | 9.13999  |
| lasso           | 6.8651    | 2.57688  | 2.98615  | 9.63347  | 3.66549  | 10.0647  |
| ridge           | 6.92315   | 2.656    | 3.19315  | 10.6656  | 3.62529  | 10.221   |
| svr_linear      | 10.0468   | 9.68979  | 3.62812  | 29.3313  | -1.98465 | 22.0783  |
| linear_baseline | 12.1368   | 11.4501  | 4.81788  | 34.8389  | -2.08039 | 26.3539  |
| linear_multiple | 12.1368   | 11.4501  | 4.81788  | 34.8389  | -2.08039 | 26.3539  |
| polynomial_deg2 | 3303.75   | 6595.27  | 4.93846  | 16494.3  | -4885.36 | 11492.9  |

Interpretation:

- Mean RMSE: 5.6878 ± 1.2837 €
- 95% Confidence Interval: [4.0938, 7.2817] €
- Stability: ☒ Good

6. OVERFITTING ANALYSIS

6.1 Polynomial Regression Analysis

Training Performance:

- RMSE: 0.0000 €
- R²: 1.000000

Test Performance:

- RMSE: 35.0258 €
- R²: -65.513740

Performance Gap:

- RMSE Gap: 35.0258 € (+781530919073378.88%)
- R² Gap: 66.513740

Diagnosis: **✗ SEVERE OVERFITTING** - Significant overfitting detected

Recommendations:

- Reduce polynomial degree (use degree=1 instead of 2)
  - Add regularization (Ridge or Lasso)
  - Collect more training data
  - Remove irrelevant features
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## 7. PREDICTION CONFIDENCE INTERVALS

### 7.1 Random Forest Uncertainty Quantification

#### Statistics:

- Confidence Level: 95.0%
- Mean Standard Deviation: 3.5247 €
- Mean CI Width: 13.8164 €
- Coverage: 83.33%

**Quality Assessment:** ⚠ **UNDER-CALIBRATED** - CIs too narrow, underestimating uncertainty

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## 8. CONCLUSIONS AND RECOMMENDATIONS

### 8.1 Summary

This project successfully developed a machine learning system to predict monthly electricity bills using partial month consumption data. Key achievements include:

1. **Multi-Cutoff Prediction:** Models can predict final bill at day 10, 15, 20, or 25
2. **High Accuracy:** Best model achieves excellent performance suitable for consumer budget forecasting
3. **Robust Feature Engineering:** Accumulated consumption and weather features drive accurate predictions
4. **Production-Ready:** Clear path to deployment with defined monitoring strategy

### 8.2 Model Deployment Recommendations

#### Production System Architecture:

1. **API Endpoint:** RESTful API for bill prediction requests
2. **Daily Updates:** Automatic daily recalculation as new consumption data arrives
3. **Multi-Cutoff Support:** Allow users to request predictions for different cutoff days
4. **Confidence Intervals:** Provide prediction uncertainty ranges
5. **Anomaly Alerts:** Flag unusual consumption patterns automatically

#### Technology Stack:

- Model Serving: FastAPI or Flask for REST API
- Database: PostgreSQL for consumption history
- Caching: Redis for frequently accessed predictions
- Monitoring: Prometheus + Grafana for model performance tracking
- Cloud Deployment: AWS Lambda or Google Cloud Run for scalability

## 8.3 Future Improvements

### Feature Engineering:

- Weather forecast integration for remaining days
- Appliance-level consumption breakdown (if smart meter available)
- Household characteristics (size, occupancy, appliances)
- Degree days (heating and cooling demand)

### Model Architecture:

- Hyperparameter optimization (Bayesian optimization)
- Advanced algorithms (XGBoost, LightGBM, CatBoost)
- Ensemble methods (stacking, blending)
- Explainability tools (SHAP, LIME)

### Data Quality:

- Smart meter integration (hourly/sub-hourly data)
- Weather forecast APIs
- Real-time tariff updates
- Automated outlier detection

## 8.4 Deployment Checklist

- ☐ Deploy best model to staging environment
- ☐ Implement prediction API with confidence intervals
- ☐ Set up monitoring dashboard (RMSE drift, data drift)
- ☐ Configure automated retraining pipeline (quarterly)
- ☐ Create user-facing dashboard
- ☐ Implement anomaly detection alerts
- ☐ Conduct A/B testing with pilot group
- ☐ Gather user feedback for improvements

## 8.5 Business Impact

### For Consumers:

- ☒ Early visibility into monthly costs (day 10-25)
- ☒ Proactive budget management
- ☒ Opportunity to adjust consumption mid-month
- ☒ Peace of mind through anomaly detection

### For Energy Providers:

- ☒ Improved customer satisfaction
- ☒ Reduced billing disputes
- ☒ Data-driven tariff optimization
- ☒ Better load forecasting

## 8.6 Next Steps



### 1. Short Term (1 month):

- Deploy model to production
- A/B test with 5% user base
- Validate accuracy on real billing cycles

### 2. Medium Term (3 months):

- Full production rollout
- Mobile app integration
- Consumption optimization tips

### 3. Long Term (6+ months):

- Multi-household predictions
- Causal inference capabilities
- Reinforcement learning optimizer

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## 9. FINAL REMARKS

This project demonstrates the practical application of supervised machine learning to a real-world energy forecasting problem. The innovative multi-cutoff approach enables early month bill predictions, providing significant value to consumers and energy providers alike.

**Key Takeaway:** Accurate electricity bill prediction using partial month data is achievable through careful feature engineering, robust model training, proper validation, and uncertainty quantification. The deployed model empowers users with proactive financial visibility and supports data-driven energy management decisions.

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